

Individual Contribution Report

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Title: Enhancement of Advanced Analytical Features in the Streamlit Web Application

Introduction:

This Streamlit web application analyzes annual trading statistics of companies listed on the Colombo Stock Exchange from 2006 to 2022. While the previous group analysis focused on a limited number of selected companies, my individual contribution extends the analytical depth and scope of the application by introducing advanced data analysis techniques and broader data coverage.

Description of Individual Contribution

1. Expansion of Dataset Coverage

In the previous group activity, the analysis was limited to four selected companies based on the highest turnover within 20 GICS industry groups. However, in this application, I expanded the analysis to include **all companies in the dataset**, enabling a more comprehensive and realistic view of the market.

2. Introduction of Advanced Analytical Techniques

a) Regression Analysis

I implemented regression analysis to examine relationships between financial variables. This helps identify whether variables have positive or negative relationships and supports data-driven insights.

b) Correlation Heatmap

A correlation heatmap was introduced to visualize relationships among all variables. This allows users to easily identify strong correlations, such as the close relationship between turnover, trade volume, and share volume.

c) K-Means Clustering

K-means clustering was applied to group data points based on features like closing price, turnover, trade volume, and volatility. This helps identify patterns and categorize companies into different market behavior groups.

This time series plot shows how the selected company's closing price changes over time. An upward trend indicates long-term growth and increasing investor confidence, while downward trends suggest declining performance. Fluctuations in the line highlight periods of market volatility, which may be influenced by economic conditions or company-specific events.